

Data Analytics (DA)

Module-End -5

Module End Assignment: Python For Data Analysis

Social Media Engagement Analytics

Project Overview

This project analyzed the **social_media_engagement_5000.csv** dataset using Python libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Plotly**. The objective was to clean, transform, visualize, and analyze social media engagement data to identify user behavior patterns, content performance, and engagement trends.

Data Cleaning & Preparation

The dataset was successfully imported into Google Colab and processed for analysis. Several preprocessing techniques were applied:

- Missing values were identified and handled using median and mode imputation.
 - Duplicate rows were removed to improve data quality.
 - Date columns were converted into datetime format for time-based analysis.
 - Incorrect categorical labels such as gender values were standardized.
 - Unrealistic negative engagement values in likes, comments, and shares were corrected.
 - Additional features such as:
 - engagement_score
 - hashtag_count
 - log-transformed likeswere created for deeper analysis.
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Key Analytical Findings

1. Content Performance

- Certain **post types** generated significantly higher engagement scores compared to others.
- Entertainment and promotional categories showed stronger audience interaction.
- Posts with higher impressions generally resulted in higher likes and shares, showing a positive correlation between visibility and engagement.

Best Performing Metrics:

- High engagement posts typically had:
 - More impressions
 - More shares
 - Higher watch time
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2. User Trends

- Engagement varied across different age groups.
 - Verified accounts generally showed higher engagement rates compared to non-verified users.
 - Users with higher follower counts tended to receive more likes and comments.
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3. Behavioral Insights

- Peak engagement and impressions occurred during specific posting hours, indicating the importance of posting time.
 - Mobile users contributed significantly higher watch time compared to some other device types.
 - Posts containing hashtags generally showed better reach and engagement.
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4. Sentiment Analysis

- Positive sentiment posts achieved the highest engagement scores overall.
 - Neutral posts showed moderate engagement.
 - Negative sentiment posts generally had lower interaction levels but occasionally generated discussion through comments.
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Statistical Analysis Summary

Descriptive statistics were calculated for:

- Likes
- Comments
- Shares
- Watch Time
- Engagement Rate

- Followers

The analysis included:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Percentiles
- Skewness and Kurtosis

The distributions indicated that engagement metrics were slightly right-skewed, meaning a smaller number of posts achieved very high engagement.

Visualization Summary

More than 8 visualizations were created using:

- Matplotlib
- Seaborn
- Plotly

Charts included:

- Scatter plots
- Line charts
- Bar charts
- Pie charts
- Histograms
- Box plots
- Heatmaps
- Pair plots
- Interactive Plotly charts

These visualizations helped identify:

- Correlations between impressions and likes
- Daily engagement trends
- Category performance

- Sentiment impact
 - Device-based behavioral patterns
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Conclusion

The project successfully demonstrated how Python-based data analytics techniques can be used to derive meaningful business insights from social media engagement data. The findings show that:

- Content type,
- Posting time,
- Sentiment,
- Device usage,
- and audience reach

all strongly influence social media performance and engagement outcomes.